Payal

B.Tech in Chemical Engineering
Indian Institute of Technology Roorkee

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Education

Degree	Institute	Year
B.Tech.	Indian Institute of Technology, Roorkee	2026
Class XII, CBSE	New Happy School	2022
Class X, CBSE	New Happy School	2020

Publications

Post-Audit Mirage | NeurIPS 2026 Workshop VerifyAgents | **Accepted Poster** *Sept 26*

Post-Audit Mirage: Identifiability Limits of Offline Agent Verification, with Aman Behera

- Designed **benchmark** for Audit-View Non-Identifiability Theorem, proving e**rror-abstention tradeoff** for offline agent verifiers
- Built matched **safe-harmful world pairs** (272 pairs across three environments) with an IdentifiedRangeMonitor, showing that verifier evaluation must test both **calculation validity** and **evidential sufficiency.**

Internships & Experience

Tredence Analytics | Data Engineer *June 26 - Present*

- Configured **Genie One in Databricks** for manufacturing analysis, for **production-ready KPI retrieval** through metric views
- Designed retrieval architecture, **routing queries** by complexity across Genie One, metric views and fact tables in parallel, with **fallback models** to reduce hallucination and keep latency low
- Improved answer accuracy from **70% to 95.4%**, refining Genie One's instructions adding few-shot question-to-SQL examples

Center for Sustainable Energy, IIT Roorkee | Research Intern *May 25 - July 25*

- Developed **regression models** based prediction system on **5,000+ material records** to identify anti-icing of road pavements
- Benchmarked **Random Forest and XGBoost** using 5-fold cross-validation & hyperparameter tuning reducing **RMSE by 18%**
- Ranked materials by predicted melting score and applied **SHAP-based explainability** to identify the key driving feature

GlueX Protocol, Remote | Software Intern *Jan 25 - March 25*

- Integrated **10+ DeFi protocols** into a liquidity aggregator, expanding supported protocol coverage by **3x** across the aggregator
- Resolved **8+ routing logic defects**, improving **swap quote accuracy to 99.9%** across all protocol integrations in the aggregator
- Improved **test coverage** from **93% to 98%** across all protocol integrations, keeping them stable, scalable and regression-safe

Projects

Hinglish Caller Bench[Open source(v0.2)] | Code *Oct 26*

- Built an **open-source evaluation harness** PyPI library that tests customer-service **AI agents** using simulated Indian callers
- Designed a split-provider architecture with per-provider token-bucket **rate limiting** and append-only **JSONL** checkpointing
- Grades **task completion, tool correctness and language fit** orthogonally; reports **confidence intervals and pass^k** metrics

Personal Knowledge OS | Code *April 26*

- Dockerised **RAG system** with stateful **LangGraph** autonomous agent routing between personal-memory and live-web tools
- Built the **async ingestion backend** (FastAPI + Celery/Redis) with parent-child chunking, sentence-transformer embeddings
- Formed **hybrid retriever** that fuses Qdrant dense-vector and Neo4j entity-graph results via **RRF** and then **Cross-Encoder**

Readthedocs | Code | Live Demo *Sep 25*

- Built an **LangChain RAG pipeline** over AWS documentation, reducing hallucinations via chunk-grounded retrieval
- Integrated **HuggingFace embeddings, ChromaDB and FastAPI** to serve scalable, source-grounded LLM inference APIs
- Built **Eval harness** scoring Recall@1/3/5, MRR, inline-citation accuracy, & abstention, with a separate judge model (Mistral)

Skills and Coursework

Languages: Python (primary), C/C++, SQL **Tools & Cloud:** Git/GitHub, Azure, Claude Code
AI Evaluation & Reliability: Agent evaluation harnesses, LLM-as-judge design and probe sets, pass^k reliability, confidence intervals and anytime-valid sequential testing, benchmark and matched-pair experiment design, failure analysis
LLM & Agent Systems: LangChain, LangGraph, RAG pipelines, MCP, voice agents (STT/TTS), Hugging Face, fine-tuning
ML & Data: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, statistical modeling, time-series analysis, computer vision

Extra Curricular

- **Winner, HiLabs AiQuest'25**: Built the Roster Email Parser for extracting structured information from healthcare emails.
- **Amazon ML Summer School'25**: Selected for advanced training in ML and AI from Amazon industry experts.